# MAT201 Linear Algebra · Midterm 2025

Time: 90 minutes. Total: 44 points.
Show all work. Answers without justification receive no credit.
Write vectors as column vectors and give each eigenvector together with its eigenvalue.

**Question 1.** Compute the determinant of A = [[3, 2], [1, 4]]. Is A invertible? Justify your answer. **[6 pts]**

**Question 2.** Let A = [[4, 1], [2, 3]]. (a) Find the characteristic polynomial of A. (b) Find the eigenvalues of A. (c) Find one eigenvector for each eigenvalue. **[10 pts]**

**Question 3.** Is v = (1, 1) an eigenvector of B = [[2, 1], [1, 2]]? If so, give its eigenvalue. Show the multiplication. **[6 pts]**

**Question 4.** Let C = [[1, 2], [2, 1]]. Find the eigenvalues of C and check your answer using the trace and the determinant of C. **[8 pts]**

**Question 5.** Explain geometrically what it means for a 2×2 matrix to have an eigenvalue equal to 0. What does this say about its determinant? **[6 pts]**

**Question 6.** Show that the rotation matrix R = [[0, −1], [1, 0]] has no real eigenvalues. Interpret your answer geometrically. **[8 pts]**

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